SETTLEXPERT CRM — MASTER TECHNICAL BLUEPRINT & SYSTEM SPECIFICATION

> **Document Type:** Reverse-Engineered Master System Specification & Technical Data Dictionary

> **Source Project:** SettleXpert CRM (settl-expert-admin v1.0.0)

> **Inspection Date:** October 2026

> **Classification:** Confidential / Internal Engineering Blueprint

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01 — CRM Overview

SettleXpert CRM is an end-to-end debt settlement, creditor harassment mitigation, legal defense, and installment fee collections platform. The system coordinates the entire client lifecycle across four core functional tiers:

1. **Sales & Telecalling (Employee CRM):** Inbound/outbound lead acquisition, debt qualification, follow-ups, and conversion.
2. **Operations & Case Management (Manager Center):** Lead quota distribution, employee calling audits, payment verification approvals, and escalation triage.
3. **Legal Advocacy (Advocate Portal):** Legal representation, anti-harassment legal notices, Section 138 NI Act counter-replies, Demand Notices (`DN-XXXX`), and Speed Post dispatch tracking.
4. **Corporate Governance & Finance (Super Admin):** Staff directory, Aadhaar KYC verification, department/team hierarchy, fee plans, collections ledger, and security audit trails.

Core Business Flow

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LEAD (Inquiry)
  ↓ [Calling / Qualification / Follow-ups]
CONVERTED CLIENT (360° Profile & Creditor Portfolio)
  ↓ [Annexure A Lenders & Fee Schedule]
CONSULTANCY AGREEMENT (Contract Execution: AGR-XXXX)
  ↓ [Installment Plan & Payment Proof Submission]
PENDING VERIFICATION (Manager Review Queue)
  ↓ [Verify & Approve]
VERIFIED PAYMENT & ACTIVE LEGAL DEFENSE
  ↓ [Anti-Harassment Notices & Settlement Negotiations]
CASE CLOSURE & NO DUE CERTIFICATE (NDC)
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02 — Technology Stack

- **Frontend Core:** React 19.2.8, React DOM 19.2.8, TypeScript 7.0.2, Vite 8.2.2.
- **Styling & UI:** Tailwind CSS 3.4.17, PostCSS 8.5.26, Autoprefixer 10.5.4, Lucide React (v1.33.0 icons), `clsx`, `tailwind-merge`.
- **Backend Core:** Node.js (CommonJS), Express.js 5.2.1, Morgan logger, CORS, dotenv.
- **Database Architecture (Dual Engine):**
- *Local / Low Latency:* SQLite 3 via `better-sqlite3` v13.0.3 (`server/data/settlexpert.db`).
- *Cloud Production:* Supabase PostgreSQL via `@supabase/supabase-js` v2.112.4.
- *Synchronization:* Two-way sync queue (`server/db/syncQueue.js`), schema deployer (`deploy_schema.js`), pull/push scripts.
- **Authentication & Security:** JSON Web Tokens (`jsonwebtoken` v9.0.3), password hashing via `bcryptjs` v3.0.3, account lockout thresholds (5 failed attempts), IP and User-Agent capture in immutable audit tables.
- **Data Processing:** `xlsx` v0.18.5 (Excel parsing and generation), `uuid` v14.0.2 (ID generation).

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03 — Complete Module List

1. **Authentication & Session Security** (`/login`, `/me`)
2. **Executive Admin Dashboard** (`/dashboard`)
3. **Staff Directory & KYC Management** (`/organization/staff-directory`)
4. **Managers Management** (`/organization/managers`)
5. **Employees Management** (`/organization/employees`)
6. **Departments Management** (`/organization/departments`)
7. **Teams Management** (`/organization/teams`)
8. **Manager Types & Classifications** (`/organization/manager-types`)
9. **Roles & Granular Permissions Matrix** (`/organization/roles-permissions`)
10. **Leads Pipeline & Calling Desk** (`/crm/leads`)
11. **Lead Import & Smart Distribution Engine** (`/crm/lead-import-distribution`)
12. **Clients Directory & Master Ledger** (`/crm/clients`)
13. **Client 360° Unified Workspace (19 Integrated Sub-Tabs)**
14. **Client Onboarding Form Engine** (`/crm/onboarding-form`)
15. **Agreements & Contracts Engine** (`/crm/agreements`)
16. **Advocates Directory & Empanelment** (`/advocates`)
17. **Advocate Legal Portal & Case Management** (`/advocate-portal`)
18. **Legal Notices Engine (Anti-Harassment & Sec 138)** (`/advocate-portal/legal-notices`)
19. **Demand Notices Engine** (`/advocate-portal/demand-notices`)
20. **Finance & Payment Verification Desk** (`/finance/payments`)

21. **Master Fee Plans** (`/finance/fee-plans`)
22. **Operations & Cross-Functional Tasks** (`/operations/tasks`)
23. **Follow-Ups Master Calendar** (`/operations/follow-ups`)
24. **Document Repository Locker** (`/operations/documents`)
25. **Escalations & Grievance Desk** (`/operations/escalations`)
26. **Reports & Executive Analytics** (`/reports`)
27. **Security & Audit Logs Trail** (`/audit`)
28. **Admin Platform Settings** (`/settings`)

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04 — Complete Page & Screen Inventory

Screen Name	File Component Path	Primary URL / Navigation Key	Primary Roles
Login Screen	`src/components/auth/LoginView.tsx`	`/` (Unauthenticated)	All Users
Admin Master Dashboard	`src/components/dashboard/AdminDashboard.tsx`	`dashboard`	Admin
Staff Directory & KYC	`src/components/staff/StaffDirectory.tsx`	`staff-directory`	Admin
Managers Management	`src/components/organization/ManagersView.tsx`	`managers`	Admin
Employees Management	`src/components/organization/EmployeesView.tsx`	`employees`	Admin
Departments	`src/components/organization/DepartmentsView.tsx`	`departments`	Admin
Teams	`src/components/organization/TeamsView.tsx`	`teams`	Admin
Manager Types	`src/components/organization/ManagerTypesView.tsx`	`manager-types`	Admin
Roles & Permissions	`src/components/organization/RolesPermissionsView.tsx`	`roles-permissions`	Admin
Leads Pipeline	`src/components/crm/LeadsView.tsx`	`leads`	Admin, Manager
Lead Import & Distribution	`src/components/leads/LeadImportDistributionView.tsx`	`lead-import-distribution`	Admin, Manager
Clients Master List	`src/components/crm/ClientsView.tsx`	`clients`	Admin

Screen Name	File Component Path	Primary URL / Navigation Key	Primary Roles
Client 360° Profile	`src/components/crm/ClientDetailsUnifiedView.tsx`	(Sub-view / Modal)	Admin, Manager, Employee
Client Onboarding Form	`src/components/crm/ClientOnboardingMainView.tsx`	`onboarding-form`	Admin, Employee
Agreements Master List	`src/components/crm/AgreementsView.tsx`	`agreements`	Admin
Printable Legal Agreement	`src/components/employee/AgreementDocumentView.tsx`	(Document Preview Mode)	All Roles
Advocates Directory	`src/components/advocates/AdvocatesView.tsx`	`advocates`	Admin
Payments Desk	`src/components/finance/PaymentsView.tsx`	`payments`	Admin, Finance
Fee Plans	`src/components/finance/FeePlansView.tsx`	`fee-plans`	Admin
Operations (Tasks/Docs)	`src/components/operations/OperationsView.tsx`	`tasks`, `follow-ups`, `documents`, `escalations`	Admin
Reports & Analytics	`src/components/reports/ReportsView.tsx`	`reports`	Admin
Security Audit Logs	`src/components/audit/AuditLogsView.tsx`	`audit`	Admin
Platform Settings	`src/components/settings/AdminSettingsView.tsx`	`settings`	Admin
Manager Workspace	`src/components/manager/ManagerWorkspace.tsx`	(Dedicated Manager Root)	Manager
Manager Dashboard	`src/components/manager/ManagerDashboardView.tsx`	`dashboard`	Manager
Manager Team View	`src/components/manager/ManagerTeamView.tsx`	`team`	Manager
Manager Leads View	`src/components/manager/ManagerLeadsView.tsx`	`leads`	Manager
Manager Clients View	`src/components/manager/ManagerClientsView.tsx`	`clients`	Manager

Screen Name	File Component Path	Primary URL / Navigation Key	Primary Roles
Manager Verification Queue	`src/components/manager/ManagerPaymentsVerificationView.tsx`	`verification`	Manager
Manager Payment Management	`src/components/manager/ManagerPaymentManagementView.tsx`	`payments`	Manager
Manager Agreements View	`src/components/manager/ManagerAgreementsView.tsx`	`agreements`	Manager
Manager Follow-Ups View	`src/components/manager/ManagerFollowUpsView.tsx`	`follow-ups`	Manager
Manager Tasks View	`src/components/manager/ManagerTasksView.tsx`	`tasks`	Manager
Manager Advocates View	`src/components/manager/ManagerAdvocatesView.tsx`	`advocates`	Manager
Manager Activity Trail	`src/components/manager/ManagerActivityView.tsx`	`activity`	Manager
Employee Workspace	`src/components/employee/EmployeeWorkspace.tsx`	(Dedicated Employee Root)	Employee
Employee Dashboard	`src/components/employee/EmployeeDashboardView.tsx`	`dashboard`	Employee
Employee Leads Desk	`src/components/employee/EmployeeLeadsView.tsx`	`leads`	Employee
Employee Clients Desk	`src/components/employee/EmployeeClientsView.tsx`	`clients`	Employee
Employee Agreements Desk	`src/components/employee/EmployeeAgreementsView.tsx`	`agreements`	Employee
Advocate Workspace	`src/components/advocate/AdvocateWorkspace.tsx`	(Dedicated Advocate Root)	Advocate
Advocate Dashboard	`src/components/advocate/AdvocateDashboardView.tsx`	`dashboard`	Advocate
Advocate Cases Directory	`src/components/advocate/AdvocateCasesView.tsx`	`cases`	Advocate

Screen Name	File Component Path	Primary URL / Navigation Key	Primary Roles
Advocate Legal Notices	`src/components/advocate/AdvocateLegalNoticesView.tsx`	`legal_notices`	Advocate
Legal Notice Document	`src/components/advocate/LegalNoticeDocumentView.tsx`	(Printable Notice Mode)	Advocate
Advocate Demand Notices	`src/components/advocate/AdvocateDemandNoticesView.tsx`	`demand_notices`	Advocate
Demand Notice Document	`src/components/advocate/DemandNoticeDocumentView.tsx`	(Printable Demand Mode)	Advocate
Advocate Tasks	`src/components/advocate/AdvocateTasksView.tsx`	`tasks`	Advocate

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05 — Role-Wise Complete Page Inventory

1. Super Admin Role

- `dashboard` (Executive KPIs, revenue stats, active pipeline, advocate summaries)
- `staff-directory` (All staff, role filters, Aadhaar KYC verification modal)
- `managers` (Manager roster, department heads, manager types, assign team)
- `employees` (Consultants/telecallers, reporting managers, passwords)
- `departments` (Department codes, head manager, team counts)
- `teams` (Team lead assignment, employee roster)
- `manager-types` (Sales, Legal, Operations, Finance permissions)
- `roles-permissions` (Fine-grained capability matrix)
- `leads` (Full company leads pipeline, bulk actions, reassignment)
- `lead-import-distribution` (Excel/CSV upload, deduplication, auto round-robin)
- `clients` (Full client directory, 360 unified view, status toggles)
- `agreements` (Company-wide contracts, execution tracking)
- `onboarding-form` (Full multi-step customer onboarding)
- `advocates` (Bar Council roster, active caseloads)
- `payments` (All payments, Verification queue, Collections ledger)
- `fee-plans` (Master pricing tiers and installment definitions)
- `tasks`, `follow-ups`, `documents`, `escalations` (Central Operations)
- `reports` (Performance, targets, collections)
- `audit` (Immutable system logs)

- `settings` (Company info, security lockout parameters, webhooks)
- *1-Click Workspace Preview Mode:* Can preview Manager, Employee, and Advocate workspaces at will via header buttons.

2. Manager Role

- `dashboard` (Team KPI cards, calling metrics, team target progress)
- `team` (Assigned consultants, active status, reset employee password)
- `leads` (Team lead pool, status filters, manual reassignment to agents)
- `lead-import-distribution` (Import and distribute leads to team members)
- `clients` (Team client directory, 360 access, advocate assignment)
- `onboarding-form` (Onboard customer directly)
- `agreements` (Agreements generated by team consultants)
- `verification` (Audit employee submitted payments, Verify & Approve, Reject with mandatory reason)
- `payments` (Team monthly collection schedule and client EMI tracker)
- `follow-ups` (Calendar and list of team scheduled callbacks)
- `tasks` (Assign operational tasks to consultants)
- `advocates` (View assigned legal counsel per client)
- `activity` (Audit trail of team actions)

3. Employee (Consultant / Telecaller) Role

- `dashboard` (Personal lead pipeline counters, conversion rate, monthly collections)
- `leads` (Personal assigned leads, calling statuses, 18-field edit lead, follow-up drawer, convert lead)
- `onboarding-form` (Multi-section onboarding form)
- `clients` (Personal converted clients, 360 profile, creditor lenders management)
- `agreements` (Agreement creation, loan details annexure, legal template generator)
- *Payments Tab (Inside Client 360):* "Submit Payment" action (Amount > 0, Payment Date, Mode, UTR, mandatory Screenshot Proof). Cannot self-verify.

4. Advocate (Legal Counsel) Role

- `dashboard` (Assigned active cases, closed cases, total debt under protection)
- `cases` (Client debt summary, creditor loan accounts, legal task logging)
- `legal_notices` (Draft, Dispatched, Delivered notices; Speed Post tracking; printable advocate letterhead notices)
- `demand_notices` (Formal counter-demand notices to aggressive collection agencies)
- `tasks` (Court hearing dates, reply filing deadlines, police station representation)

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06 — Permission Matrix

Module	Super Admin	Manager	Employee	Advocate
Users / Staff	View, Create, Edit, Delete	View Team	View Self	View Self
KYC Verification	View, Approve, Reject	View	None	View Self
Leads	View All, Create, Edit, Delete, Assign	View Team, Assign	View Assigned, Create, Edit	None
Lead Conversion	Execute	Execute	Execute	None
Clients	View All, Edit, Delete, Assign	View Team, Assign Adv	View Assigned, Edit	View Assigned Cases
Lenders / Creditors	View, Add, Delete	View, Add, Delete	View, Add, Delete	View
Agreements	View All, Generate, Delete	View Team, Generate	View Assigned, Generate	View Linked
Payment Submission	Submit	Submit	Submit (Pending Only)	None
Payment Verification	Approve, Reject	Approve, Reject	**FORBIDDEN (403)**	None
Legal Notices	View All	View	None	Draft, Dispatch, Print
Fee Plans	Create, Edit, Toggle	View	View	None
Audit Logs	View All, Filter	View Team Activity	None	None
Settings	Full Access	None	None	None

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07 — Lead Module Specification

1. Data Fields

Field Name	Display Label	Data Type	Required	Default Value	Notes
`id`	Internal ID	String (UUID)	Yes	Auto `lead-xxxxxxxx`	Primary Key
`lead_number`	Lead #	String	Yes	Auto `LEAD-0001`	Sequential Unique Code
`name`	Full Name	String	Yes	—	Customer Name
`phone`	Phone Number	String	Yes	—	10-digit, deduplicated

Field Name	Display Label	Data Type	Required	Default Value	Notes
`email`	Email Address	String	No	null	Deduplicated (NOCASE)
`city`	City	String	No	null	Client City
`state`	State	String	No	null	Client State
`address`	Full Address	String	No	null	Street address
`total_debt`	Total Debt	Numeric	No	0.00	Total loan/card dues
`monthly_income`	Monthly Income	Numeric	No	0.00	Net monthly income
`service_needed`	Service Needed	String	No	'Debt Settlement'	e.g. Personal Loan, Card
`paying_emis`	Paying EMIs	Enum/String	No	'Yes'	'Yes', 'No', 'Irregular'
`harassment_calls`	Harassment Calls	Enum/String	No	'No'	'No', 'Mild', 'Severe'
`employment_status`	Employment Status	String	No	'Employed'	'Employed', 'Self-Employed'
`employment_type`	Employment Type	String	No	'Salaried'	'Salaried', 'Business'
`settlement_needed`	Settlement Needed	String	No	'Yes'	'Yes', 'No'
`consultation_timing`	Consultation Time	String	No	'10:00 AM - 12:00 PM'	Slot preferences
`credit_card_dues`	Credit Card Dues	Numeric	No	0.00	Breakdown
`personal_loan_dues`	Personal Loan Dues	Numeric	No	0.00	Breakdown
`service_fee`	Service Fee	Numeric	No	0.00	Estimated SX fee
`bank_name`	Bank / Creditor	String	No	null	Primary lender
`status`	Lead Status	Enum	Yes	'new'	Pipeline status
`employee_id`	Assigned Agent	String (FK)	No	Logged in user	References `users.id`
`manager_id`	Assigned Manager	String (FK)	No	Agent's manager	References `users.id`
`created_by`	Creator Name	String	No	Logged in name	Audit field
`created_at`	Created At	Timestamp	Yes	CURRENT_TIMESTAMP	Filter anchor

Field Name	Display Label	Data Type	Required	Default Value	Notes
`updated_at`	Updated At	Timestamp	Yes	CURRENT_TIMESTAMP	Last activity

2. Lead Status Values

- `all`: Meta-tab representing all non-converted, non-deleted leads.
- `new`: Freshly imported or uncontacted lead.
- `connected`: Telecaller initiated call and customer answered.
- `contacted`: Initial discussion regarding debt relief completed.
- `interested`: Customer confirmed interest in debt settlement terms.
- `follow_up`: Scheduled callback booked for a specific date and time.
- `converted`: Successfully enrolled and transitioned to a Client record.
- `not_interested` / `dropped`: Customer declined services or loans ineligible.
- `deleted`: Soft-deleted record.

3. Follow-Up Action Drawer

- **Past Follow-Up History Table:** Displays historical interactions with `Date & Time`, `Agent Name`, `Call Status`, `Interest Level`, `Remarks`.
- **New Follow-Up Form:**
- `Call Status` Dropdown: `Connected`, `Not Reachable`, `Busy`, `Switch Off`, `Wrong Number`.
- `Interest Level` Dropdown: `High`, `Medium`, `Low`.
- `Final Status` Dropdown: `Interested`, `Follow_up`, `Converted`, `Dropped`.
- `Next Follow-Up Date & Time`: Integrated Date & Time Picker.
- `Call Remark`: Textarea for detailed call synopsis.

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08 — Lead to Client Conversion Engine

The conversion from a Lead to a Client is a traceable transaction that guarantees no data re-entry and prevents duplicate records.

Conversion Logic & Field Mapping

Lead Source Field	Client Target Field	Transformation / Logic
`lead.id`	`client.lead_id`	Preserved as Foreign Key link
Auto-generated	`client.client_number`	Sequential `CL-0001`, `CL-0002`...
`lead.name`	`client.name`	Direct copy
`lead.phone`	`client.phone`	Direct copy (Unique index)
`lead.email`	`client.email`	Direct copy

Lead Source Field	Client Target Field	Transformation / Logic
`lead.city`	`client.city`	Direct copy
`lead.address`	`client.address`	Direct copy
`lead.state`	`client.state`	Direct copy
`lead.employment_status`	`client.employment_status`	Direct copy
`lead.employment_type`	`client.employment_type`	Direct copy
`lead.total_debt`	`client.total_debt`	Direct copy
`lead.monthly_income`	`client.monthly_income`	Direct copy
`lead.credit_card_dues`	`client.credit_card_dues`	Direct copy
`lead.personal_loan_dues`	`client.personal_loan_dues`	Direct copy
`lead.bank_name` / `service`	`client.loan_type`	`Credit Card & Loans (${bank_name})`
`lead.paying_emis`	`client.paying_emis`	Direct copy
`lead.harassment_calls`	`client.harassment_calls`	Direct copy
`lead.settlement_needed`	`client.settlement_needed`	Direct copy
`lead.consultation_timing`	`client.consultation_timing`	Direct copy
`lead.total_debt`	`client.settlement_target`	Auto-calculated: `ROUND(total_debt * 0.45)` (45% target)
`lead.service_fee`	`client.sx_fee`	Auto-populated: `lead.service_fee	25000`
Execution Date	`client.fees_date`	`CURRENT_DATE`
Initial State	`client.fees_status`	`'Pending'`
Initial State	`client.total_received`	`0.00`
Initial State	`client.pending_amount`	Equals `client.sx_fee`
Initial State	`client.status`	`'active'`
Initial State	`client.case_status`	`'active'`
`lead.employee_id`	`client.employee_id`	Direct ownership transfer
`lead.manager_id`	`client.manager_id`	Direct hierarchy transfer

Automated Secondary Operations on Conversion:

1. **Initial Creditor Creation:** If `lead.bank_name` is present, a record is automatically inserted into `lenders`:

- `bank_name = lead.bank_name`
- `loan_type = 'Credit Card / Personal Loan'`

- `balance = lead.total_debt`
- `status = 'Defaulted'`

2. **Consultancy Agreement Creation:** Automatically executes `ensureAgreementForClient()`:
- Generates sequential `AGR-XXXX`.
- Links `client_id`.
- Populates Annexure A with the initial bank lender.
- Populates Annexure B with total consultancy fee and 6-month monthly installment breakdown (`ROUND(totalFee / 6)`).

3. **Status Update:** `UPDATE leads SET status = 'converted', updated_at = CURRENT_TIMESTAMP WHERE id = ?`.

4. **Duplicate Prevention:** If a client already exists with `lead_id = lead.id` or `phone = lead.phone`, the system does NOT create a duplicate record; it re-links the existing client.

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09 — Client 360° Unified Workspace (All 19 Sub-Tabs)

When opening a client record in `ClientDetailsUnifiedView.tsx`, the following 19 sub-tabs are rendered:

Tab 1: `onboarding-form` (Client Onboarding Form)

- **Purpose:** Full digital onboarding form containing KYC documents, proof of income, household budget, creditor loan list, and legal consent checkboxes.

Tab 2: `info` (Profile & Debt Overview)

- **Purpose:** Client summary master card. Displays contact particulars, total enrolled debt, 45% settlement target, Sx service fee, assigned consultant, and quick-update Case Status dropdown (`active`, `closed`, `on_hold`, `dropped`).

Tab 3: `lenders` (Creditors & Loan Accounts)

- **Purpose:** Bank-by-bank ledger of unsecured debts.
- **Table Columns:** `Bank Name`, `Loan Type` (Personal Loan / Credit Card), `Balance (₹)`, `Default Date`, `Status` (Defaulted / NPA), `Statement Link`.
- **Actions:** `+ Add More` modal (Bank Name, Type, Balance), `Remove` (delete lender with confirmation dialog).

Tab 4: `income-emi` (Financial Assessment)

- **Purpose:** Affordability index calculation.
- **Fields:** Net Monthly Income vs Fixed Expenses (`House Rent`, `Groceries`, `School Fees`, `Utilities`, `Medical`).
- **Calculation:** Auto-computes **Net Disposable Surplus** available for the monthly settlement escrow pool.

Tab 5: `emi` (Monthly EMI Schedule Tracker)

- **Purpose:** Month-by-month installment schedule from Month 1 to Month 12+.

- **Columns:** `Month #`, `Due Date`, `Expected Amount`, `Received Amount`, `Balance`, `Status` (`Paid`, `Pending`, `Overdue`).

Tab 6: `bank` (Bank & NACH / Auto-Debit Controls)

- **Purpose:** Client bank account details (`Bank Name`, `A/C Number`, `IFSC Code`, `Branch`).
- **Tool: NACH / E-Mandate Disablement Advisory** toggle to guide the client on cancelling auto-debits to eliminate bank bounce penalty charges.

Tab 7: `documents` (Client Document Locker)

- **Purpose:** Secure digital file storage.
- **Categories:** Aadhaar Front/Back, PAN Card, Last 6 Months Bank Statements, Loan Sanction Letters, Creditor Demand Notices.
- **Actions:** Drag-and-drop file upload, document preview modal, download, delete.

Tab 8: `boarding-call` (Welcome / Boarding Call Verification)

- **Purpose:** Operational QA check.
- **Form:** Checklist confirming client was briefed on settlement duration, creditor harassment response protocol, and fee payment rules. Uploads call audio recording and consultant signoff notes.

Tab 9: `conversation` (Communication Logs)

- **Purpose:** Interactive timestamped activity thread.
- **Fields:** Category selector (`Call`, `WhatsApp`, `Email`, `Internal Staff Note`), free-text note area, `Add Note` button. Full conversational audit trail.

Tab 10: `client-action` (Action Triggers & Follow-up Scheduling)

- **Purpose:** Cross-departmental task assignment.
- **Fields:** `Next Call Date & Time`, `Assign Task To` (Legal Advocate / Accounts / Manager), `Priority` (Urgent, High, Normal), `Action Instructions`.

Tab 11: `case-closure` (Case Closure & NDC)

- **Purpose:** Final settlement milestone desk.
- **Actions:** Upload Bank Settlement Letter, upload official Bank No Due Certificate (NDC), record final fee clearance, and update master client status to `closed`.

Tab 12: `client-attribute` (Risk & Severity Attributes)

- **Purpose:** Severity tags: `Severe Harassment`, `Police Complaint Threat`, `High Net-Worth`, `Arbitration Notice Issued`, `Cooperative`.

Tab 13: `notices` (Bank Legal Notices)

- **Purpose:** Central registry of legal notices received by the client from banks: Section 138 NI Act, Section 25 Payment & Settlement Act, Arbitration summons, Lok Adalat notices.

Tab 14: `sales-handover` (Consultant Handover)

- **Purpose:** Handover protocol from Sales Telecaller to Operations Team. Captures special client commitments, emotional state, payment frequency promises, and handover signoff.

Tab 15: `payment-history` (Receipts & Payment Ledger)

- **Purpose:** Complete canonical transaction history for this client.
- **Table Columns:** `Receipt #`, `Payment Date`, `Amount (₹)`, `Payment Mode`, `Transaction / UTR ID`, `Proof Document Link`, `Status Badge` (Pending Verification / Verified / Rejected), `Verified / Rejected By`, `Rejection Reason`.
- **Action:** `+ Submit Payment` button (opens manual payment submission modal).

Tab 16: `escalation-form` (Harassment & Grievance Escalation)

- **Purpose:** Urgent ticket submission to Manager or Legal Lead.
- **Fields:** `Subject`, `Escalation Category` (Recovery agent home visit, third-party abusive calls, employer contact), `Urgency` (Critical, High), `Incident Description`.

Tab 17: `financial-form` (Distress Balance Sheet)

- **Purpose:** Complete debt-to-income (DTI) evaluation, total liquid assets vs unsecured debt liabilities, affordability score.

Tab 18: `sales-feedback-form` (CSAT & Quality Check)

- **Purpose:** 5-star rating matrix on sales transparency, fee clarity, consultant professionalism, and client comments.

Tab 19: `history` (Audit Trail Timeline)

- **Purpose:** Full chronological audit log of all status transitions, field edits, advocate assignments, and payment verifications for this specific client profile.

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10 — Agreements Module & Legal Contract Specification

Agreement Form Sections (`EmployeeAgreementsView.tsx`)

1. Client Details Section

- `Name *`: Text input (Client Full Name).
- `PIN Number / PAN`: Text input (`placeholder="e.g. FGWPK0024B"`).
- `Date of Birth`: Date picker (`type="date"`).
- `Phone *`: Tel input (10 digits).
- `Email *`: Email input.
- `Start Date`: Date picker.
- `Address`: Textarea (2 rows).

2. Loan Details (Lenders Table)

- Dynamic table with rows:
- `Lender Name`: Input box (`placeholder="Lender Name"`).
- `Loan Type`: Input box (`placeholder="e.g. Personal Loan, Credit Card"`).
- `Loan Amount`: Numeric input (`font-mono`, e.g. `150000`).
- `Remove`: Red button (`Trash` icon), disabled if only 1 lender exists.
- `+ Add More`: Blue button, appends a new lender row.
- `Total Loan`: Disabled read-only field calculating `SUM(balance)` live.

3. Agreement Details Section

- `Agreement Duration`: Dropdown / Text input (e.g. `1 Month`, `2 Months`, `4 Months`, `6 Months`, `12 Months`, `Lifetime`). Automatically updates monthly installment fee.
- `Consultancy Fees (Monthly)`: Auto-calculated or manual override (e.g. `Rs. 7,500`).
- `Resolution Duration`: Text input (e.g. `6 Months`).
- `Prepared By`: Text input (e.g. `Dhruv Consultant`).
- `Executed Date`: Date picker.
- `Monthly Income (₹)`: Numeric input.

4. Official 12-Clause Agreement Contract Body

The system dynamically injects variables into the official legal contract text:

- **Title:** `CONSULTANCY AGREEMENT`
- **Execution Date:** Dynamically populated `${executed_date}`.
- **First Party:** M/s. SettleXpert LLP, CB-201, Naraina Vihar, Ring Road, New Delhi, Delhi, India.
- **Second Party:** `${clientName}`, residing at `${clientAddress}`, DOB: `${clientDob}`, Mobile: `${clientPhone}`, Email: `${clientEmail}`.
- **Clause 1 (Purpose):** Appointment of company for financial consultancy and debt resolution advisory.
- **Clause 2 (Scope of Services):** Evaluating liabilities, formulating resolution strategies, communicating/negotiating with lenders, drafting replies/representations, anti-harassment guidance.
- **Clause 3 (Company Obligations):** Diligence, confidentiality, regular client updates.
- **Clause 4 (Client Obligations):** Providing truthful information, timely fee payments, prompt notification of lender communications, non-borrowing during tenure.
- **Clause 5 (Client Acknowledgements):** Advisory only, no guarantee of waiver/settlement, timeline depends on lenders.
- **Clause 6 (Fees & Payment Terms):** Non-refundable consultancy fee, strict installment schedule, service suspension on default.
- **Clause 7 (Term & Termination):** Tenure duration, termination upon notice or non-payment.
- **Clause 8 (Limitation of Liability):** Company not liable for lender legal actions or recovery agency conduct.
- **Clause 9 (Confidentiality):** Non-disclosure of sensitive financial data.
- **Clause 10 (Grievance Redressal):** Escalation procedure before legal recourse.

- **Clause 11 (Governing Law & Jurisdiction):** Courts of New Delhi, India.
- **Clause 12 (Entire Agreement):** Supersedes prior discussions.
- **Annexure A:** Auto-formatted dynamic table of enrolled lenders and loan balances.
- **Annexure B:** Fee breakdown: Total Agreed Fee (`₹${totalFee}`), Monthly Installment (`₹${monthlyFee}`), Tenure (`${tenure}`).
- **Execution Blocks:** Dual signature blocks for Authorized Signatory (SettleXpert LLP) and Client Signature.

---

11 — Payment & Verification Workflow (No Razorpay)

The payment flow is strictly manual payment submission followed by managerial verification. Razorpay has been removed from this operational workflow.

Step-by-Step Payment Lifecycle

```
1. EMPLOYEE SUBMISSION:
   Employee opens Client 360 → Tab 15 (Payment History) → Clicks "+ Submit Payment"
   Fields entered: Amount Paid (> 0), Payment Date, Payment Mode (UPI, IMPS, NEFT, Cash),
   Transaction/UTR ID (Required for non-cash), Payment Proof Screenshot (MANDATORY), Remarks.
   ↓
2. PENDING VERIFICATION:
   Payment inserted into `payments` table with `status = 'pending_verification'`.
   Audit log recorded: "Payment Submitted".
   Employee CANNOT self-verify or approve payment.
   ↓
3. MANAGER REVIEW QUEUE:
   Payment appears in Manager Work Center (`ManagerPaymentsVerificationView.tsx`)
   and Finance Desk (`PaymentsView.tsx`).
   Manager reviews: Client Name, Agreement #, Amount, Payment Date, Payment Mode, UTR, Proof Screenshot.
   ↓
4. VERIFICATION / REJECTION:
   - ACTION A: VERIFY & APPROVE
     - Sets `status = 'verified'`, `verified_by = req.user.name`, `verified_at = CURRENT_TIMESTAMP`.
     - Atomically recalculates client balance:
       `total_received = SUM(amount WHERE status = 'verified')`
       `pending_amount = MAX(0, sx_fee - total_received)`
       `fees_status = total_received >= sx_fee ? 'Paid' : 'Partially Paid'`
     - Syncs to client profile and finance reports.
   - ACTION B: REJECT PAYMENT
     - Rejection reason is MANDATORY (e.g. UTR mismatch, amount not credited, invalid proof).
     - Sets `status = 'rejected'`, `rejected_by = req.user.name`, `rejected_at = CURRENT_TIMESTAMP`, `rej
     - Employee sees "PAYMENT REJECTED" with the exact reason and can resubmit with corrected proof.
```

---

12 — Database Schema (All 26 Tables)

Table 1: `users`

- `id` TEXT PRIMARY KEY

- `name` TEXT NOT NULL
- `email` TEXT NOT NULL UNIQUE
- `phone` TEXT
- `password_hash` TEXT NOT NULL
- `role` TEXT NOT NULL CHECK(role IN ('admin', 'manager', 'employee', 'advocate'))
- `manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `department_id` TEXT REFERENCES departments(id) ON DELETE SET NULL
- `team_id` TEXT REFERENCES teams(id) ON DELETE SET NULL
- `manager_type_id` TEXT REFERENCES manager_types(id) ON DELETE SET NULL
- `emp_or_mgr_id` TEXT UNIQUE
- `profile_image` TEXT
- `joining_date` TEXT
- `id_type` TEXT
- `id_front` TEXT
- `id_back` TEXT
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `failed_login_attempts` INTEGER DEFAULT 0
- `locked_until` DATETIME
- `last_failed_login_at` DATETIME
- `last_login_at` DATETIME
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 2: `departments`

- `id` TEXT PRIMARY KEY
- `name` TEXT NOT NULL
- `code` TEXT NOT NULL UNIQUE
- `description` TEXT
- `head_manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 3: `teams`

- `id` TEXT PRIMARY KEY
- `name` TEXT NOT NULL
- `code` TEXT NOT NULL UNIQUE
- `department_id` TEXT REFERENCES departments(id) ON DELETE SET NULL

- `manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 4: `manager_types`

- `id` TEXT PRIMARY KEY
- `name` TEXT NOT NULL
- `code` TEXT NOT NULL UNIQUE
- `description` TEXT
- `permissions_json` TEXT DEFAULT '{}'
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 5: `user_permissions`

- `id` TEXT PRIMARY KEY
- `user_id` TEXT NOT NULL REFERENCES users(id) ON DELETE CASCADE
- `module` TEXT NOT NULL
- `can_view` INTEGER NOT NULL DEFAULT 1
- `can_create` INTEGER NOT NULL DEFAULT 0
- `can_edit` INTEGER NOT NULL DEFAULT 0
- `can_delete` INTEGER NOT NULL DEFAULT 0
- `can_assign` INTEGER NOT NULL DEFAULT 0
- `can_verify` INTEGER NOT NULL DEFAULT 0
- UNIQUE(user_id, module)

Table 6: `advocates`

- `id` TEXT PRIMARY KEY
- `name` TEXT NOT NULL
- `advocate_id` TEXT NOT NULL UNIQUE
- `mobile` TEXT NOT NULL
- `email` TEXT NOT NULL UNIQUE
- `registration_number` TEXT NOT NULL UNIQUE
- `specialization` TEXT NOT NULL
- `address` TEXT
- `profile_image` TEXT
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `notes` TEXT

- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 7: `fee_plans`

- `id` TEXT PRIMARY KEY
- `name` TEXT NOT NULL
- `duration` TEXT NOT NULL
- `default_fee` NUMERIC(12, 2) NOT NULL
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'inactive'))
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 8: `audit_logs`

- `id` TEXT PRIMARY KEY
- `user_id` TEXT
- `user_name` TEXT
- `role` TEXT
- `action` TEXT NOT NULL
- `module` TEXT NOT NULL
- `record_id` TEXT
- `details_json` TEXT
- `ip_address` TEXT
- `user_agent` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 9: `leads`

- `id` TEXT PRIMARY KEY
- `lead_number` TEXT UNIQUE NOT NULL
- `name` TEXT NOT NULL
- `phone` TEXT NOT NULL
- `email` TEXT
- `city` TEXT
- `state` TEXT
- `address` TEXT
- `total_debt` REAL DEFAULT 0
- `monthly_income` REAL DEFAULT 0
- `service_needed` TEXT
- `paying_emis` TEXT DEFAULT 'Yes'
- `harassment_calls` TEXT DEFAULT 'No'
- `employment_status` TEXT DEFAULT 'Employed'
- `employment_type` TEXT DEFAULT 'Salaried'

- `settlement_needed` TEXT DEFAULT 'Yes'
- `consultation_timing` TEXT
- `credit_card_dues` REAL DEFAULT 0
- `personal_loan_dues` REAL DEFAULT 0
- `service_fee` REAL DEFAULT 0
- `bank_name` TEXT
- `status` TEXT NOT NULL DEFAULT 'new'
- `manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `employee_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `department_id` TEXT REFERENCES departments(id) ON DELETE SET NULL
- `created_by` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 10: `follow_ups`

- `id` TEXT PRIMARY KEY
- `lead_id` TEXT NOT NULL REFERENCES leads(id) ON DELETE CASCADE
- `user_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `user_name` TEXT
- `call_status` TEXT NOT NULL
- `interested_level` TEXT
- `final_status` TEXT NOT NULL
- `remark` TEXT
- `next_follow_up_date` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 11: `clients`

- `id` TEXT PRIMARY KEY
- `client_number` TEXT UNIQUE NOT NULL
- `lead_id` TEXT REFERENCES leads(id) ON DELETE SET NULL
- `name` TEXT NOT NULL
- `email` TEXT
- `phone` TEXT NOT NULL
- `city` TEXT
- `address` TEXT
- `state` TEXT
- `employment_status` TEXT DEFAULT 'Employed'
- `employment_type` TEXT DEFAULT 'Salaried'

- `total_debt` REAL DEFAULT 0
- `monthly_income` REAL DEFAULT 0
- `credit_card_dues` REAL DEFAULT 0
- `personal_loan_dues` REAL DEFAULT 0
- `loan_type` TEXT DEFAULT 'Credit Card & Personal Loan'
- `paying_emis` TEXT DEFAULT 'Yes'
- `harassment_calls` TEXT DEFAULT 'No'
- `settlement_needed` TEXT DEFAULT 'Yes'
- `consultation_timing` TEXT
- `settlement_target` REAL DEFAULT 0
- `sx_fee` REAL DEFAULT 0
- `fees_date` TEXT
- `fees_status` TEXT DEFAULT 'Pending'
- `pending_amount` REAL DEFAULT 0
- `total_received` REAL DEFAULT 0
- `status` TEXT NOT NULL DEFAULT 'active'
- `case_status` TEXT DEFAULT 'active'
- `notes` TEXT
- `employee_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `advocate_id` TEXT REFERENCES advocates(id) ON DELETE SET NULL
- `fee_plan_id` TEXT REFERENCES fee_plans(id) ON DELETE SET NULL
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 12: `lenders`

- `id` TEXT PRIMARY KEY
- `client_id` TEXT NOT NULL REFERENCES clients(id) ON DELETE CASCADE
- `bank_name` TEXT NOT NULL
- `loan_type` TEXT
- `balance` REAL DEFAULT 0
- `default_date` TEXT
- `status` TEXT DEFAULT 'Defaulted'
- `lender_email` TEXT
- `pdf_url` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 13: `agreements`

- `id` TEXT PRIMARY KEY
- `agreement_number` TEXT UNIQUE NOT NULL
- `client_id` TEXT REFERENCES clients(id) ON DELETE CASCADE
- `fee_plan_id` TEXT REFERENCES fee_plans(id) ON DELETE SET NULL
- `name` TEXT
- `phone` TEXT
- `email` TEXT
- `address` TEXT
- `pin_number` TEXT
- `dob` TEXT
- `start_date` TEXT
- `end_date` TEXT
- `total_fee` REAL NOT NULL
- `monthly_fee` REAL DEFAULT 0
- `monthly_income` REAL DEFAULT 0
- `resolution_duration` TEXT
- `prepared_by` TEXT
- `executed_date` TEXT
- `status` TEXT NOT NULL DEFAULT 'draft' CHECK(status IN ('draft', 'sent', 'signed', 'active', 'completed', 'cancelled'))
- `agreement_body` TEXT
- `created_by` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 14: `payments`

- `id` TEXT PRIMARY KEY
- `receipt_number` TEXT UNIQUE
- `client_id` TEXT REFERENCES clients(id) ON DELETE CASCADE
- `agreement_id` TEXT REFERENCES agreements(id) ON DELETE SET NULL
- `amount` REAL NOT NULL
- `payment_method` TEXT NOT NULL
- `payment_mode` TEXT
- `transaction_id` TEXT
- `proof_document` TEXT
- `remarks` TEXT

- `status` TEXT NOT NULL DEFAULT 'pending_verification' CHECK(status IN ('pending_verification', 'verified', 'rejected'))
- `submitted_by` TEXT REFERENCES users(id) ON DELETE SET NULL
- `submitted_by_name` TEXT
- `submitted_at` DATETIME
- `verified_by` TEXT
- `verified_at` DATETIME
- `rejected_by` TEXT REFERENCES users(id) ON DELETE SET NULL
- `rejected_by_name` TEXT
- `rejected_at` DATETIME
- `rejection_reason` TEXT
- `payment_date` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 15: `monthly_payment_records`

- `id` TEXT PRIMARY KEY
- `client_id` TEXT NOT NULL REFERENCES clients(id) ON DELETE CASCADE
- `agreement_id` TEXT REFERENCES agreements(id) ON DELETE SET NULL
- `month_number` INTEGER NOT NULL
- `due_date` TEXT NOT NULL
- `expected_amount` REAL NOT NULL DEFAULT 0
- `received_amount` REAL NOT NULL DEFAULT 0
- `payment_date` TEXT
- `payment_status` TEXT NOT NULL DEFAULT 'Pending' CHECK(payment_status IN ('Pending', 'Partially Paid', 'Paid', 'Overdue'))
- `remarks` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 16: `monthly_payment_history`

- `id` TEXT PRIMARY KEY
- `monthly_record_id` TEXT NOT NULL REFERENCES monthly_payment_records(id) ON DELETE CASCADE
- `client_id` TEXT NOT NULL REFERENCES clients(id) ON DELETE CASCADE
- `agreement_id` TEXT REFERENCES agreements(id) ON DELETE SET NULL
- `month_number` INTEGER NOT NULL
- `payment_date` TEXT NOT NULL
- `amount_received` REAL NOT NULL
- `cumulative_received` REAL NOT NULL

- `payment_status` TEXT NOT NULL
- `remarks` TEXT
- `updated_by_id` TEXT
- `updated_by_name` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 17: `payment_due_notifications`

- `id` TEXT PRIMARY KEY
- `client_id` TEXT NOT NULL REFERENCES clients(id) ON DELETE CASCADE
- `agreement_id` TEXT REFERENCES agreements(id) ON DELETE SET NULL
- `monthly_record_id` TEXT REFERENCES monthly_payment_records(id) ON DELETE CASCADE
- `month_number` INTEGER NOT NULL
- `notification_type` TEXT NOT NULL CHECK(notification_type IN ('PAYMENT_DUE_TODAY', 'PARTIAL_PAYMENT', 'PAYMENT_OVERDUE', 'PAYMENT_RECEIVED'))
- `title` TEXT NOT NULL
- `message` TEXT NOT NULL
- `due_date` TEXT NOT NULL
- `expected_amount` REAL NOT NULL DEFAULT 8000
- `received_amount` REAL NOT NULL DEFAULT 0
- `pending_amount` REAL NOT NULL DEFAULT 0
- `days_overdue` INTEGER NOT NULL DEFAULT 0
- `status` TEXT NOT NULL DEFAULT 'active' CHECK(status IN ('active', 'resolved'))
- `is_read` INTEGER NOT NULL DEFAULT 0
- `employee_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 18: `tasks`

- `id` TEXT PRIMARY KEY
- `title` TEXT NOT NULL
- `description` TEXT
- `module` TEXT
- `priority` TEXT DEFAULT 'medium' CHECK(priority IN ('low', 'medium', 'high', 'urgent'))
- `due_date` TEXT
- `status` TEXT NOT NULL DEFAULT 'pending' CHECK(status IN ('pending', 'in_progress', 'completed', 'cancelled'))
- `assigned_to` TEXT REFERENCES users(id) ON DELETE SET NULL
- `created_by` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 19: `lead_imports`

- `id` TEXT PRIMARY KEY
- `import_number` TEXT UNIQUE NOT NULL
- `file_name` TEXT
- `uploaded_by` TEXT REFERENCES users(id)
- `uploaded_by_name` TEXT
- `uploaded_by_role` TEXT
- `total_rows` INTEGER DEFAULT 0
- `valid_leads` INTEGER DEFAULT 0
- `duplicate_leads` INTEGER DEFAULT 0
- `invalid_rows` INTEGER DEFAULT 0
- `imported_leads` INTEGER DEFAULT 0
- `distribution_status` TEXT DEFAULT 'pending'
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 20: `lead_distributions`

- `id` TEXT PRIMARY KEY
- `distribution_number` TEXT UNIQUE NOT NULL
- `import_id` TEXT REFERENCES lead_imports(id) ON DELETE SET NULL
- `distributed_by` TEXT REFERENCES users(id)
- `distributed_by_name` TEXT
- `distributed_by_role` TEXT
- `distribution_mode` TEXT NOT NULL
- `assignment_method` TEXT NOT NULL
- `total_leads` INTEGER NOT NULL
- `employee_count` INTEGER NOT NULL
- `summary_json` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 21: `staff_extended_profiles`

- `user_id` TEXT PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE
- `date_of_birth` TEXT
- `gender` TEXT
- `father_name` TEXT
- `mother_name` TEXT
- `alt_phone` TEXT
- `whatsapp_number` TEXT
- `current_address` TEXT

- `permanent_address` TEXT
- `city` TEXT
- `state` TEXT
- `pin_code` TEXT
- `designation` TEXT
- `employment_status` TEXT DEFAULT 'Active'
- `staff_type` TEXT DEFAULT 'Employee'
- `reporting_manager_id` TEXT REFERENCES users(id) ON DELETE SET NULL
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 22: `staff_kyc`

- `id` TEXT PRIMARY KEY
- `user_id` TEXT NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE
- `aadhaar_number` TEXT
- `aadhaar_front_doc` TEXT
- `aadhaar_back_doc` TEXT
- `kyc_status` TEXT DEFAULT 'pending' CHECK(kyc_status IN ('pending', 'uploaded', 'verified', 'rejected'))
- `verified_by` TEXT
- `verified_at` DATETIME
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 23: `staff_advocate_details`

- `user_id` TEXT PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE
- `registration_number` TEXT
- `bar_council_state` TEXT
- `specialization` TEXT
- `years_experience` INTEGER DEFAULT 0
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 24: `advocate_assignment_history`

- `id` TEXT PRIMARY KEY
- `client_id` TEXT NOT NULL REFERENCES clients(id) ON DELETE CASCADE
- `previous_advocate_id` TEXT
- `previous_advocate_name` TEXT
- `new_advocate_id` TEXT
- `new_advocate_name` TEXT
- `assigned_by_id` TEXT

- `assigned_by_name` TEXT
- `assigned_by_role` TEXT
- `notes` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 25: `legal_notices`

- `id` TEXT PRIMARY KEY
- `notice_number` TEXT UNIQUE NOT NULL (Format: `LN-0001`)
- `client_id` TEXT REFERENCES clients(id) ON DELETE CASCADE
- `advocate_id` TEXT REFERENCES advocates(id) ON DELETE SET NULL
- `bank_name` TEXT NOT NULL
- `loan_account_no` TEXT
- `notice_type` TEXT NOT NULL (Anti-Harassment, Reply to Legal Notice, Sec 138 Reply, Arbitration Response)
- `notice_date` TEXT NOT NULL
- `speed_post_number` TEXT
- `dispatch_date` TEXT
- `delivery_date` TEXT
- `status` TEXT NOT NULL DEFAULT 'Draft' (Draft, Dispatched, Delivered)
- `notice_body` TEXT
- `created_by` TEXT
- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

Table 26: `demand_notices`

- `id` TEXT PRIMARY KEY
- `demand_number` TEXT UNIQUE NOT NULL (Format: `DN-0001`)
- `client_id` TEXT REFERENCES clients(id) ON DELETE CASCADE
- `advocate_id` TEXT REFERENCES advocates(id) ON DELETE SET NULL
- `bank_name` TEXT NOT NULL
- `loan_account_no` TEXT
- `demand_type` TEXT NOT NULL
- `notice_date` TEXT NOT NULL
- `speed_post_number` TEXT
- `dispatch_date` TEXT
- `delivery_date` TEXT
- `status` TEXT NOT NULL DEFAULT 'Draft'
- `notice_body` TEXT
- `created_by` TEXT

- `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP
- `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP

---

13 — Complete Backend API Inventory

Auth Routes (`/api/auth`)

- `POST /login` : Body `{ email, password }`. Verifies bcrypt hash, lockout check. Returns JWT token and User profile.
- `GET /me` : Returns logged in user profile, permissions, department, manager info.
- `POST /change-password` : Body `{ current_password, new_password }`.
- `POST /logout` : Audit logout event.

CRM Leads Routes (`/api/crm`)

- `GET /crm/leads` : Params `{ search, status, date, calendar_date, from_date, to_date, page, limit }`. Returns leads list, dynamic status counts, pagination.
- `GET /crm/leads/:id` : Returns single lead particulars and complete `follow_ups` history.
- `POST /crm/leads` : Body `{ name, phone, email, city, total_debt, monthly_income, service_needed, paying_emis, harassment_calls, ... }`. Validates phone/email deduplication. Generates `LEAD-XXXX`.
- `PUT /crm/leads/:id` : Updates lead information without altering original `created_at`.
- `DELETE /crm/leads/:id` : Soft deletes lead (`status = 'deleted'`).
- `POST /crm/leads/:id/follow-up` : Body `{ call_status, interested_level, final_status, remark, next_follow_up_date }`. Appends to follow-up history log.
- `POST /crm/leads/:id/convert` : Converts lead into Client (`CL-XXXX`), creates Annexure A lender, and generates agreement (`AGR-XXXX`).

CRM Clients Routes (`/api/crm`)

- `GET /crm/clients` : Params `{ search, case_status, month, date, calendar_date, from_date, to_date, page, limit }`. Returns enriched client array with total received, pending fees, duration.
- `GET /crm/clients/:id` : Returns client 360 particulars, attached `lenders`, `agreements`, and `payments`.
- `POST /crm/clients` : Directly creates a new client and generates `CL-XXXX`.
- `PUT /crm/clients/:id` : Updates client particulars, address, case status, fees.
- `DELETE /crm/clients/:id` : Removes client record and cascades dependencies.
- `POST /crm/clients/:id/lenders` : Body `{ bank_name, loan_type, balance }`. Adds creditor bank.
- `DELETE /crm/clients/:id/lenders/:lenderId` : Deletes creditor lender.
- `GET /crm/clients/:id/monthly-payments` : Returns monthly EMI breakdown records.
- `POST /crm/clients/:id/monthly-payments/generate` : Generates Month 1 to Month N payment schedule.
- `PUT /crm/monthly-payments/:recordId` : Updates specific monthly record amount/status.

CRM Agreements Routes (`/api/crm`)

- `GET /crm/agreements` : Params `{ search, status, date, calendar_date, from_date, to_date, page, limit }` .
- `GET /crm/agreements/:id` : Returns agreement details and dynamic template body.
- `POST /crm/agreements` : Body `{ client_id, total_fee, monthly_fee, resolution_duration, lenders, agreement_body, ... }` . Generates `AGR-XXXX` .
- `PUT /crm/agreements/:id` : Updates agreement terms, body, or status (`draft` , `sent` , `signed` , `active` , `completed` , `cancelled`).
- `DELETE /crm/agreements/:id` : Deletes agreement.

CRM Payments Routes (`/api/crm`)

- `GET /crm/payments` : Params `{ search, status, date, client_id, payment_mode }` . Scoped by employee/manager/admin.
- `POST /crm/payments` : Manual payment submission. Body `{ client_id, agreement_id, amount, payment_mode, transaction_id, payment_date, proof_document, remarks }` . Validates amount > 0, required proof document, generates `REC-YYYY-XXX` , sets status to `pending_verification` .
- `PATCH /crm/payments/:id/verify` : Status update by Manager/Admin. Body `{ status: 'verified' | 'rejected', notes, rejection_reason }` . Enforces 403 for employees. Atomically recalculates client balance on approval or rejection.

Manager Panel Routes (`/api/manager`)

- `GET /manager/context` : Returns manager role code, department, and team scope.
- `GET /manager/dashboard` : Returns team-wide lead metrics, consultant performance, collections.
- `GET /manager/team` : Returns list of consultants reporting to this manager.
- `POST /manager/team/:employeeId/reset-password` : Resets consultant password.
- `GET /manager/leads` : Returns leads assigned to this manager's consultants.
- `POST /manager/leads/:leadId/assign` : Body `{ new_employee_id, reason }` . Reassigns lead.
- `GET /manager/clients` : Returns clients managed by this manager's team.
- `PATCH /manager/clients/:clientId/advocate` : Body `{ advocate_id }` . Assigns advocate to client.
- `PATCH /manager/clients/:clientId/fee` : Updates client fee schedule.
- `GET /manager/payments/verification-queue` : Returns all payments with `status = 'pending_verification'` .
- `PATCH /manager/payments/:paymentId/verify` : Verifies or rejects payment with mandatory reason.
- `GET /manager/follow-ups` : Returns scheduled callbacks for team.
- `GET /manager/tasks` : Returns operational tasks.

Advocate Portal Routes (`/api/advocate-portal`)

- `GET /advocate-portal/dashboard` : Assigned cases counts, debt managed, recent cases.
- `GET /advocate-portal/cases` : Filter assigned cases by status, date, search.
- `GET /advocate-portal/cases/:id` : Detailed case file, lenders, notices, tasks.
- `POST /advocate-portal/cases/:id/tasks` : Logs legal task against client case.

- `GET /advocate-portal/legal-notices`: Returns legal notices (`LN-XXXX`), status counts (Draft, Dispatched, Delivered).
- `POST /advocate-portal/legal-notices`: Creates new legal notice.
- `PUT /advocate-portal/legal-notices/:id`: Updates notice body, Speed Post tracking number, delivery status.
- `GET /advocate-portal/demand-notices`: Returns demand notices (`DN-XXXX`).
- `POST /advocate-portal/demand-notices`: Creates new demand notice.
- `PUT /advocate-portal/demand-notices/:id`: Updates demand notice particulars.

Staff & Organization Routes

- `GET /api/staff/all`: Full staff directory with personal, KYC, and advocate details.
- `POST /api/staff/create`: Creates staff member (`users` + `staff_extended_profiles`).
- `PUT /api/staff/:id`: Updates staff personal and professional data.
- `POST /api/staff/:id/kyc`: Uploads/updates Aadhaar KYC documents and number.
- `PATCH /api/staff/:id/kyc-verify`: Body `{ kyc_status: 'verified' | 'rejected' }`.
- `GET /api/departments`: Roster of departments with head manager and employee counts.
- `POST /api/departments`: Creates department.
- `GET /api/teams`: Teams grouped under departments.
- `POST /api/teams`: Creates team and assigns team lead.
- `GET /api/manager-types`: Manager classifications and permissions JSON.
- `GET /api/fee-plans`: Master fee plans.

Lead Import & Distribution Routes (`/api/lead-import`)

- `POST /lead-import/upload`: Parses Excel/CSV, validates phone/email, deduplicates against existing records, generates summary.
- `POST /lead-import/distribute`: Distributes parsed batch to agents using Auto Round-Robin or Custom Quota mode.
- `GET /lead-import/history`: Audit log of all imports and lead allocations.

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14 — Financial Formulas & Calculation Engine

1. **Settlement Target:**

$$\text{Settlement Target} = \text{ROUND}(\text{Total Debt} \times 0.45)$$

(Standard 45% settlement goal of total enrolled debt).

2. **Total Agreement Fee:**

$$\text{Total Agreement Fee} = \text{Client Monthly Fee} \times \text{Duration in Months}$$

(Default: ₹8,000 monthly $\times$ 6 months = ₹48,000).

3. **Total Verified Received Amount (Source of Truth):**

$$\text{Total Received} = \sum \text{amount} \quad \text{WHERE } \text{status} = \text{'verified'}$$

(Pending or rejected payments are strictly excluded).

4. **Pending Client Balance:**

$$\text{Pending Balance} = \max(0, \text{Total Agreed Fee} - \text{Total Verified Received})$$

5. **Fees Status State Machine:**

$$\text{Fees Status} = \begin{cases} \text{'Paid'} & \text{if } \text{Total Received} \ge \text{Total Agreed Fee} \text{ and } \text{Fee} > 0 \\ \text{'Partially Paid'} & \text{if } \text{Total Received} > 0 \\ \text{'Pending'} & \text{otherwise} \end{cases}$$

6. **Net Disposable Surplus (Income-EMI Tab):**

$$\text{Disposable Surplus} = \text{Net Monthly Income} - (\text{Rent} + \text{Groceries} + \text{School Fees} + \text{Utilities} + \text{Medical})$$

7. **Lead Conversion Ratio:**

$$\text{Conversion Rate} = \left(\frac{\text{Total Clients}}{\text{Total Leads}}\right) \times 100$$

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15 — Legacy & Deprecated Code Audit

1. **Legacy Razorpay References (`src/components/crm/ClientDetailsUnifiedView.tsx`):**

- Lines 528 and 935 contain static placeholder text labels: `"Razorpay / UPI"`. There is no actual Razorpay SDK dependency in `package.json` or active backend webhook. In the new CRM, replace these labels with `"UPI / Net Banking"`.

2. **Legacy PHP Backend (`backend/` & `php-app/`):**

- The root contains old PHP controllers (`backend/controllers/DashboardController.php`, `DepartmentController.php`). The active production backend is completely running on Node.js / Express (`server/index.js`). The PHP files are legacy artifacts from an earlier prototype.

3. **Standalone Zip Archives (`gh.zip`, `mingit.zip`):**

- Stored in the root folder, these are portable Git binaries and not part of the CRM application runtime.

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16 — New CRM Rebuild Blueprint (Production Architecture)

When building the brand-new SettleXpert CRM from scratch, adhere to this architectural structure:

1. Unified Authentication & RBAC Service

- Implement JWT authentication with role payload: `admin`, `manager`, `employee`, `advocate`.
- Enforce backend route guards on all modifying endpoints (`POST /payments`, `PATCH /payments/:id/verify`). Never rely solely on hiding frontend buttons.

2. Single Canonical Payment Table

- Avoid splitting payments into separate tables. Use one master `payments` table with clear lifecycle statuses:
- `pending_verification`
- `verified`
- `rejected`
- Mandatory fields for submission: `amount` (> 0), `payment_date`, `payment_mode`, `transaction_id`, `proof_document` (Base64 or secure storage URL), `submitted_by`.
- Manager verification must require `rejection_reason` when rejecting and must atomically trigger client financial recalculation.

3. Traceable Lead-to-Client Pipeline

- Maintain a direct one-to-one or one-to-many foreign key relationship (`clients.lead_id -> leads.id`).
- When converting, copy all 18+ lead qualification fields directly into the new client record. Automatically generate sequential `CL-XXXX` and `AGR-XXXX`.

4. Dynamic Contract Generation

- Maintain the 12-clause legal contract template with Annexure A (Lenders) and Annexure B (Fee Schedule). Auto-compile variables into a rich-text printable view.

5. Multi-Tab Client 360 Workspace

- Retain the proven 19-tab information architecture in `ClientDetailsUnifiedView` to give legal, operations, sales, and accounts a single 360° pane of glass for each debtor.

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17 — Summary Metric Counts

- **Total Unique Functional Screens / Views:** 48
- **Total Functional Tabs:** 32 (including all 19 Client 360 sub-tabs)
- **Total Database Tables:** 26
- **Total Backend REST Endpoints:** 58
- **Total User Roles:** 4 (`admin`, `manager`, `employee`, `advocate`)
- **Total Input Forms:** 22
- **Total Master Entity Types:** 12 (`users`, `leads`, `clients`, `lenders`, `agreements`, `payments`, `tasks`, `notices`, `departments`, `teams`, `fee_plans`, `audit_logs`)